$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\big[\hat{H}_{\mathcal{S}}, \hat{\rho}_{\mathcal{S}}(t)\big]$$
$$-\int_0^\infty \mathrm{d}\tau\, \{\nu(\tau)\,[\hat{\sigma}_z, [\hat{\sigma}_z(-\tau), \hat{\rho}_{\mathcal{S}}(t)]] - \mathrm{i}\eta(\tau)\,[\hat{\sigma}_z, \{\hat{\sigma}_z(-\tau), \hat{\rho}_{\mathcal{S}}(t)\}]\}\,. \quad (5.193)$$

The kernels $\nu(\tau)$ and $\eta(\tau)$ are now given by the spin-bath expressions (5.191) and (5.192).

If we take the self-Hamiltonian of the system to be of the form $\hat{H}_{\mathcal{S}} = -\frac{1}{2}\Delta_0\hat{\sigma}_x$, as in our discussion of the spin–boson model (see Sect. 5.3.2), the